MACHINE LEARNING OPERATIONS (MLOps) · AIMLCZG523

Assignment 01 · Supplementary report

Implementation on the BITS Prayogshala Virtual Lab

The heart-disease MLOps pipeline reproduced end to end on the course lab VM: training, tests, Docker, Kubernetes (minikube) and monitoring

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Student ID	**2025AE05470**
Course	Machine Learning Operations (MLOps), AIMLCZG523
Lab	Prayogshala Virtual Lab · MLOPS (AIMLZG523) · https://prayogshala.bits-pilani.ac.in
Lab VM	Rocky Linux 9.5 · 4 vCPU · 15 GiB RAM · AWS ap-south-1 (ip-10-3-23-9)
Repository	https://github.com/callmejei/2025ae05470_assignment
Main report	`report/report.pdf` (design, EDA, modelling, CI/CD)
Date	19 October 2026

Summary

Stage on the lab VM	Result
Lint (flake8) and 14 unit tests (pytest)	**Passed**
Data download, EDA, training with MLflow tracking	**Same metrics as the original run (CV ROC-AUC 0.911, test 0.961)**
Docker image build and container smoke test	**Passed**
Docker Compose: API, Prometheus, Grafana	**Running, API healthy**
Kubernetes: manifest and Helm chart on minikube	**2/2 pods each, LoadBalancer IPs assigned**
/predict on all three endpoints	**Identical response (disease, confidence 0.9478)**
Monitoring: JSON logs, Prometheus target, Grafana dashboard	**Live data**

1. Lab environment and access

Access followed the *Prayogshala Virtual Lab Manual* (CSIS, WILP): open the portal → **Continue with BITS SSO** → log in with eLearn credentials → select **MLOPS (AIMLZG523)** → **Launch Lab** → wait for the VM to reach **RUNNING** → **Connect**, which opens an XFCE desktop in the browser.

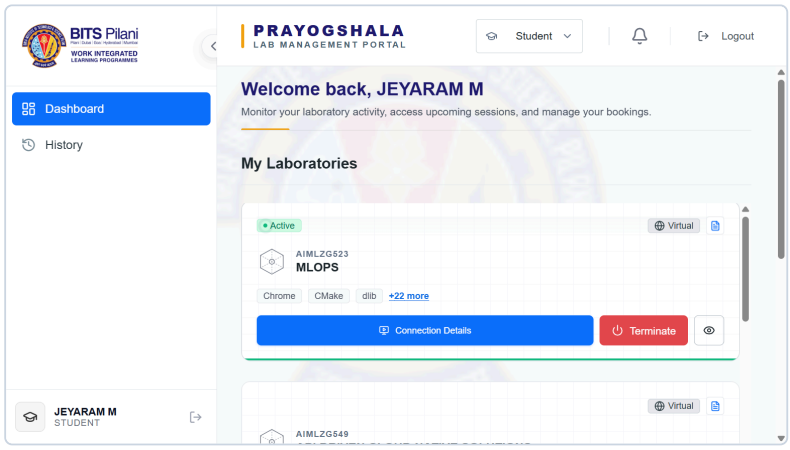


Fig 1. Prayogshala portal: MLOPS (AIMLZG523) virtual lab active for Jeyaram M

Item	Lab VM
OS / hardware	Rocky Linux 9.5 (Blue Onyx), 4 vCPU, 15 GiB RAM
Python	3.12.9 (conda base). The project's pinned versions all support 3.12, so no changes were needed.
Containers	Docker 28.0.1 (usable without sudo), Docker Compose plugin
Kubernetes	minikube v1.35.0 (Docker driver) running Kubernetes v1.32.0. kubectl comes from `minikube kubectl`; Helm v3.16.2 was downloaded by the setup script.
Persistent storage	`~/workspace` (AWS EFS). Everything outside it is lost when the VM is terminated.

Lab constraints that shaped the setup: only `~/workspace` persists; installed software is temporary; sessions end after 15 minutes idle; a stopped VM is terminated after 60 minutes. So the repository lives in `~/workspace`, the Python virtualenv lives in `~/.venvs` (fast local disk, recreated in about 75 s), and the whole stack is rebuilt by one idempotent script.

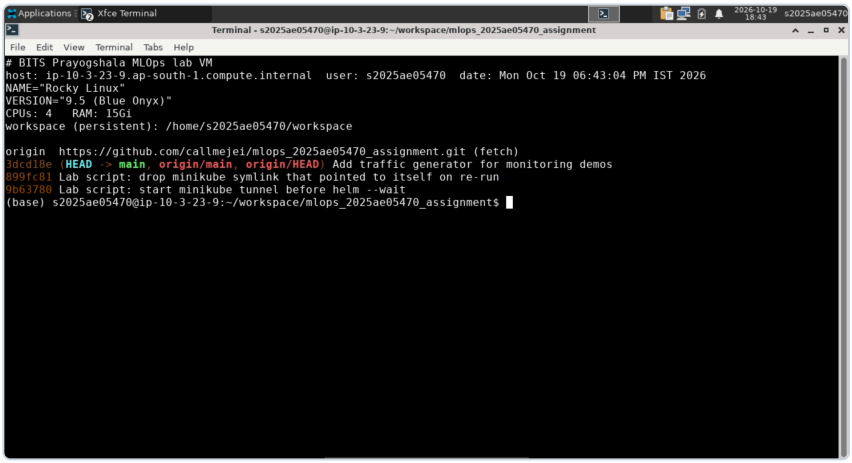


Fig 2. Lab VM identity (host, user, OS, CPU, RAM), persistent workspace, and the cloned repository at the latest commit

2. Setup procedure

Everything is driven by `scripts/lab_setup.sh`, which is in the repository. From a fresh VM, two commands reproduce the full stack:

```
cd ~/workspace
git clone https://github.com/callmejei/2025ae05470_assignment.git
bash 2025ae05470_assignment/scripts/lab_setup.sh 2>&1 | tee ~/workspace/lab_setup.log
```

#	Stage	What the script does
1	Python environment	Creates `~/.venvs/heart` and installs `requirements.txt`
2	CI pipeline	Runs `scripts/ci_local.sh`: flake8 → pytest → download & clean → EDA → train with MLflow → docker build → container smoke test. Stops at the first failure.
3	Docker Compose	Starts API :8010, Prometheus :9095, Grafana :3005 and waits for `/health`
4	Kubernetes	`minikube start`, `minikube image load`, `kubectl apply -f k8s/deployment.yaml`, rollout status
5	LoadBalancer	Runs `minikube tunnel` in the background so LoadBalancer services get an external IP
6	Helm	Installs Helm if missing, then `helm upgrade --install heart-api helm/heart-api`
7	MLflow UI	Starts the tracking UI on :5005 in the background
8	Verify	POSTs `sample_request.json` to all three endpoints

Timing (first run on a fresh VM)

Stage	Duration
Clone + Python dependencies	≈ 1 min 15 s
CI pipeline (lint, 14 tests, data, EDA, training, docker build, smoke test) + Docker Compose stack	≈ 2 min 20 s
minikube cluster start (first time) and manifest rollout	≈ 2 min 05 s
Total, fresh VM to running Kubernetes deployment	**≈ 5 min 40 s**

Issues found on the lab VM and how they were fixed (the script was then re-run on the same VM)

Symptom	Cause	Fix (committed to the repo)
Helm install failed: `context deadline exceeded`	`helm --wait` waits for LoadBalancer services to get an IP, but the tunnel only started after Helm	Start `minikube tunnel` before the Helm step
Tunnel not running on re-run: *Too many levels of symbolic links*	The script linked `~/.local/bin/minikube` to the result of `command -v minikube`, which on the second run found the link itself	Removed the link; the system minikube in `/bin` is used directly
Lab screen locked during long waits	XFCE screensaver	Disabled for the session (`xset s off`; screensaver lock off)

3. CI pipeline and reproducibility on the lab VM

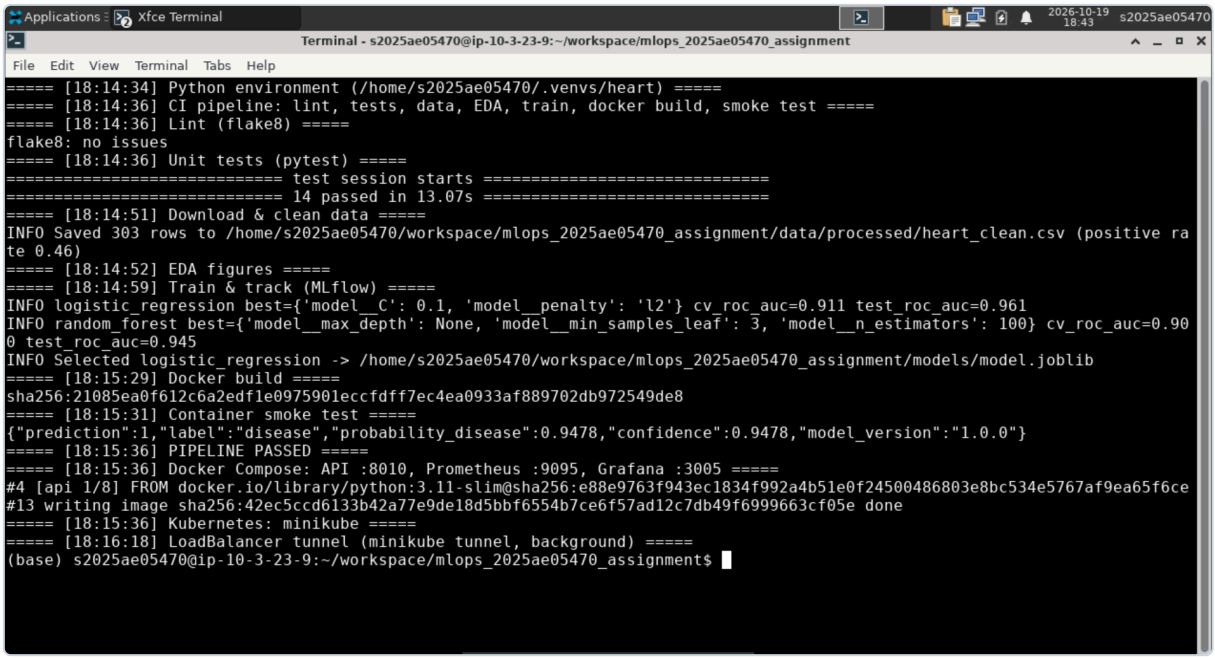


Fig 3. Pipeline log on the lab VM: lint clean, 14 tests passed, data, EDA, training, Docker build and smoke test, PIPELINE PASSED

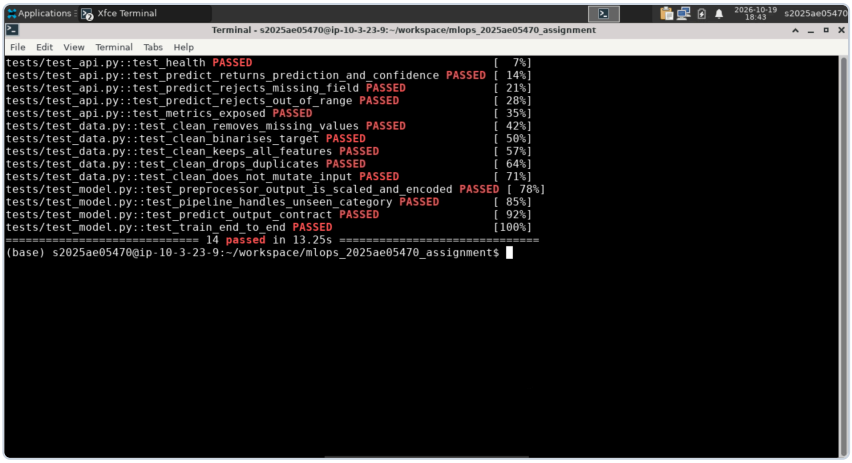


Fig 4. Unit tests on the lab VM: 14/14 passed (data cleaning, preprocessing, training, API contract and validation, metrics)

Reproducibility: original machine vs lab VM

Metric	Original run	Lab VM
Logistic Regression: best params	C = 0.1, L2	C = 0.1, L2
Logistic Regression: CV ROC-AUC / test ROC-AUC	0.911 / 0.961	0.911 / 0.961
Random Forest: CV ROC-AUC / test ROC-AUC	0.900 / 0.945	0.900 / 0.945
Selected model	logistic_regression	logistic_regression
Sample prediction (probability of disease)	0.9478	0.9478

The results match exactly despite a different OS, a different Python version (3.11 vs 3.12) and different hardware. Pinned dependencies, a fixed random seed and a single sklearn `Pipeline` make the run deterministic.

4. Containerization and Kubernetes deployment

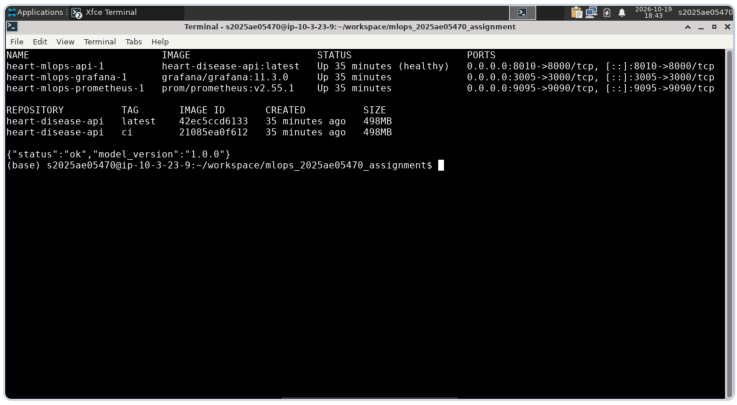


Fig 5. Docker on the lab VM: Compose stack running (API healthy), images built, /health OK

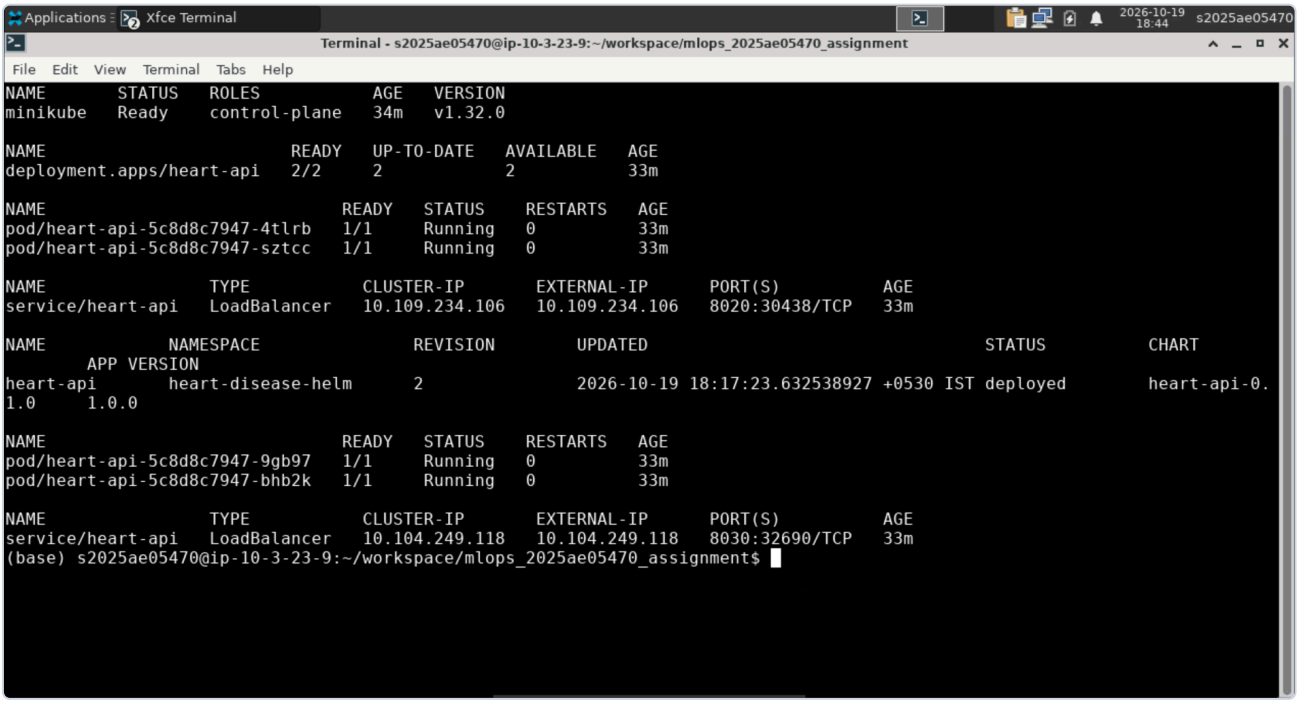


Fig 6. minikube on the lab VM: node Ready; manifest deployment 2/2 with LoadBalancer 10.109.234.106:8020; Helm release `heart-api` deployed with 2 pods and LoadBalancer 10.104.249.118:8030

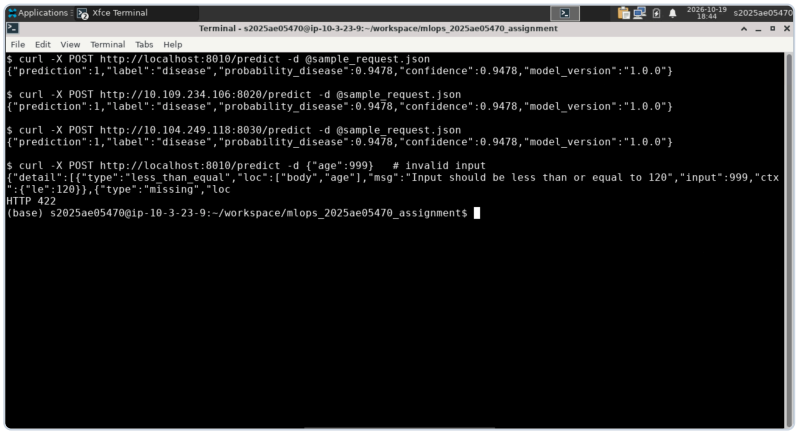


Fig 7. Endpoint verification: Docker Compose, Kubernetes manifest and Helm all return the same prediction; invalid input is rejected with HTTP 422

5. Monitoring, logging and experiment tracking

Load was generated with `scripts/send_traffic.sh`: varied patient records, with every tenth request deliberately invalid so the error-rate panel has something to show.

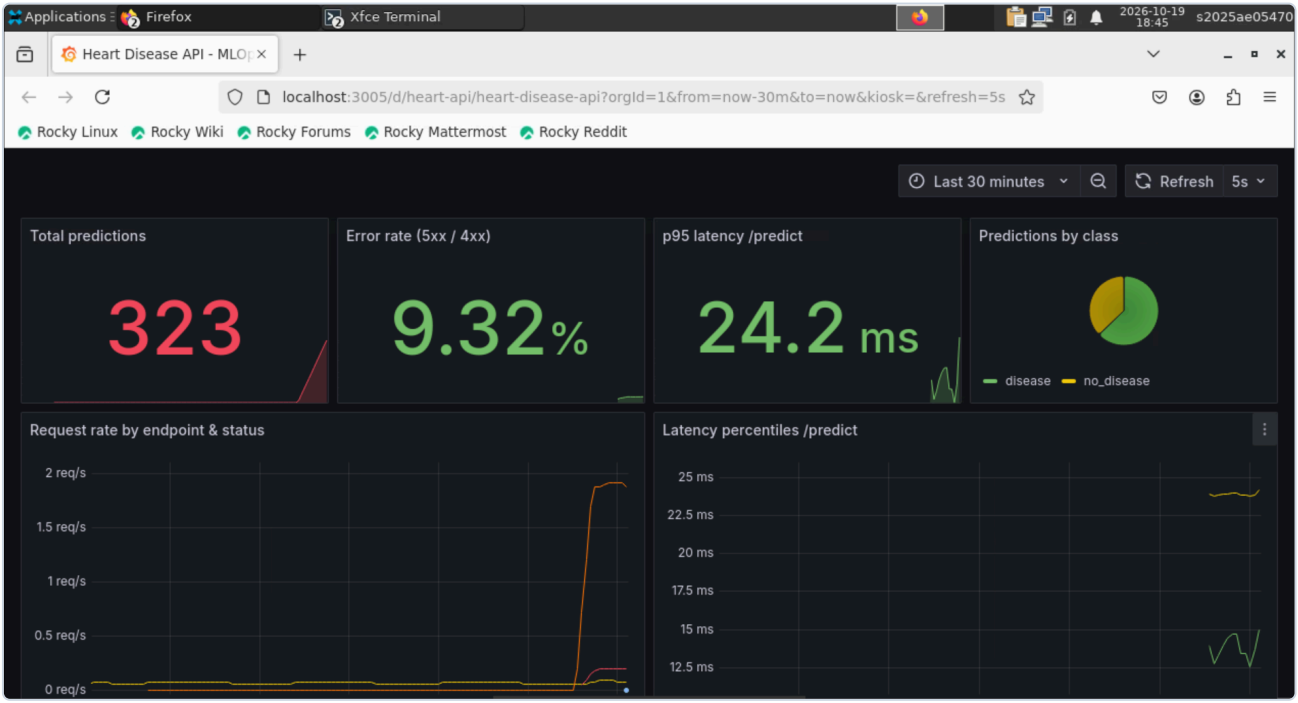


Fig 8. Grafana on the lab VM: 323 predictions, 9.3% error rate (the deliberate invalid requests), p95 latency 24 ms, class mix, request rate and latency percentiles

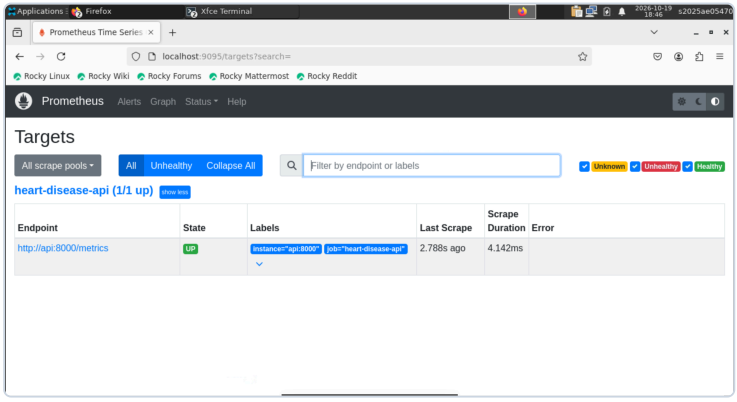


Fig 9. Prometheus target `heart-disease-api` UP, scraped every 5 s

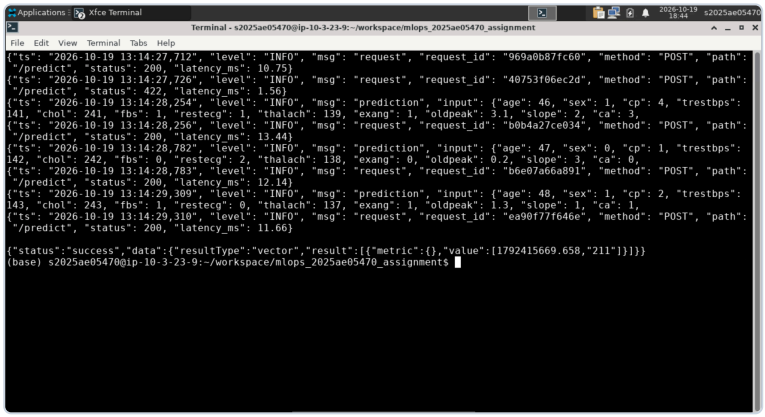


Fig 10. Structured JSON request and prediction logs (request_id, status, latency, inputs) and the Prometheus prediction counter

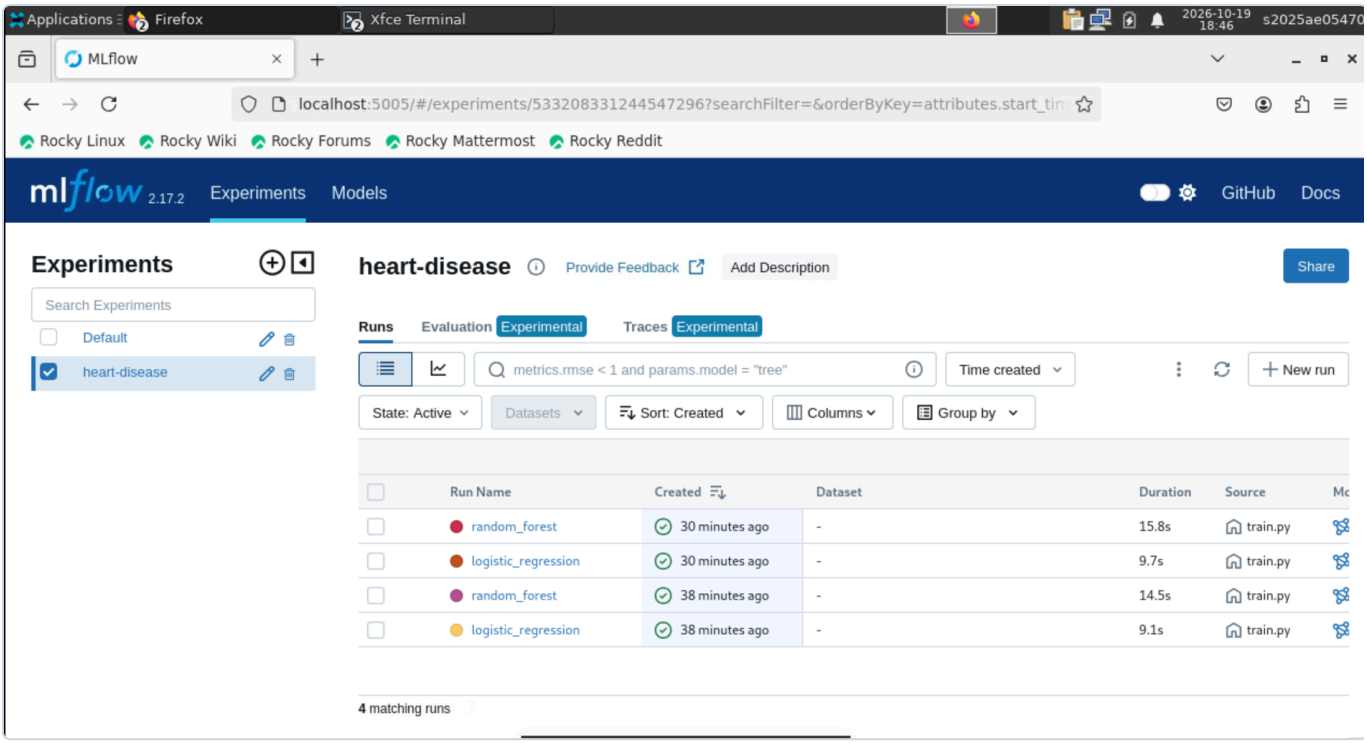


Fig 11. MLflow UI on the lab VM: experiment `heart-disease` with the tracked runs (Logistic Regression and Random Forest, two pipeline executions)

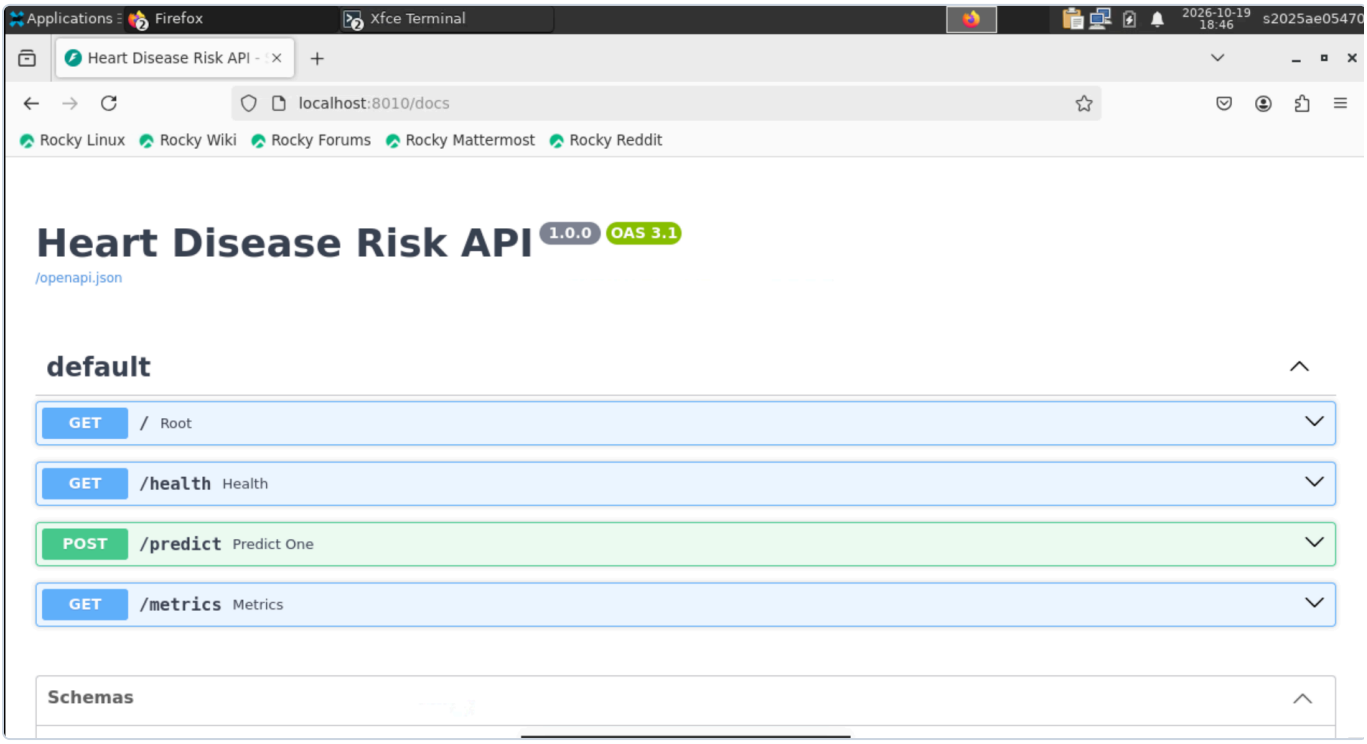


Fig 12. FastAPI Swagger UI served from the container on the lab VM

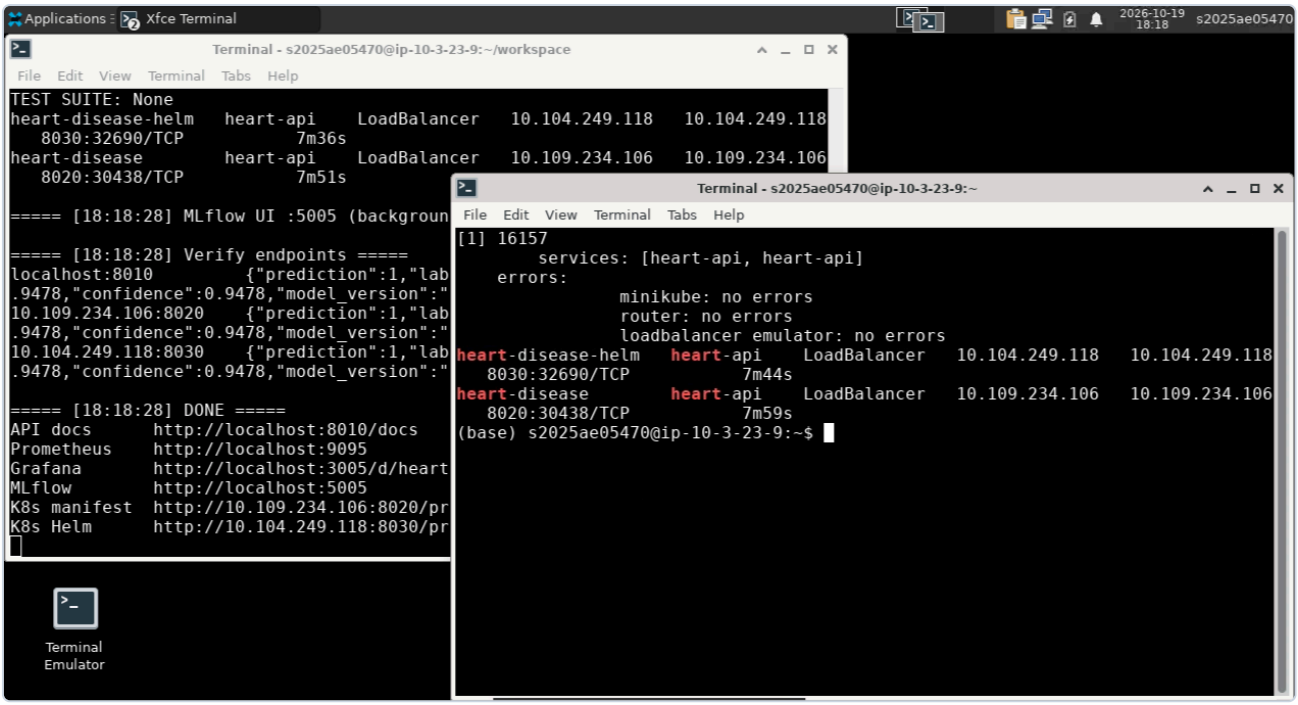


Fig 13. End of the setup script on the lab VM: LoadBalancers assigned, all endpoints verified, DONE; the second terminal shows `minikube tunnel` with no errors

6. Access summary and reproduction

Service	URL inside the lab VM
API (Swagger)	http://localhost:8010/docs
API on Kubernetes (manifest)	http://10.109.234.106:8020/predict
API on Kubernetes (Helm)	http://10.104.249.118:8030/predict
Prometheus	http://localhost:9095
Grafana (admin / admin)	http://localhost:3005/d/heart-api
MLflow	http://localhost:5005

> The LoadBalancer IPs are assigned by `minikube tunnel` and can change when a new VM is provisioned. The script prints the current addresses at the end of each run. After a VM is terminated, only `~/workspace` remains (repository, data, trained model, MLflow runs and `lab_setup.log`); re-running `scripts/lab_setup.sh` rebuilds the rest.

Files added for the lab

- `scripts/lab_setup.sh`: one-command build and deploy for the Prayogshala VM
- `scripts/send_traffic.sh`: traffic generator for the monitoring demo
- `screenshots/lab/`: evidence captured from the lab VM (Figures 1–12)